Module/Pathway/Taxon Review: `recfor_recombination` (KEGG ppu03440, Homologous Recombination) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03440` — Homologous recombination (12 primary genes; 24-gene candidate list) **Curation goal:** Species-aware module satisfiability and gene-annotation review to support manual curation. **Evidence basis:** Direct KEGG REST gene/KO mapping for KT2440 loci (verified this study) + mechanistic and *Pseudomonas*-genus literature. Strain-direct experimental phenotype from Akkaya et al. 2021.

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1. Executive Summary

The RecFOR homologous-recombination (HR) module is **satisfiable — "covered"** in *Pseudomonas putida* KT2440. Every core RecFOR presynaptic gene is encoded and confidently identified in the genome: **recF** (PP_0012), **recO** (PP_1435), **recR** (PP_4267), the strand-exchange recombinase **recA** (PP_1629), together with the shared presynaptic accessory factors **recJ** (PP_1477), **recQ** (PP_4516) and single-stranded DNA-binding protein **ssb** (PP_0485), and the full downstream branch-migration/resolution machinery **ruvA/ruvB/ruvC** (PP_1216/PP_1217/PP_1215) and **recG** (PP_5310). In addition, the alternative RecBCD presynaptic pathway is intact (**recB** PP_4673, **recC** PP_4674, **recD** PP_4672), so KT2440 carries the two canonical parallel recombination routes described for the genus *Pseudomonas*.

The single most important curation conclusion is that **the supplied 24-gene candidate list should not be treated as ground truth — it needs revision (`module_needs_revision`)**. The list is exactly the raw KEGG `map03440` membership retrieved from the KEGG REST API, and KEGG's rendering of that map deliberately shares boxes with the DNA-replication map. As a result the list is simultaneously **over-broad** and **under-complete**. It is over-broad because 12 of the 24 loci are DNA polymerase III holoenzyme subunits, DNA polymerase I, and generic exonucleases whose primary biological role is replication or excision repair, not RecFOR recombination (dnaN, polA, PP_0353, holC, dnaEA, holB, dnaQ, dnaX, PP_4768, holA, plus ssb and recJ which are mis-bucketed). It is under-complete because several bona fide genome-encoded HR/RecFOR genes are absent from the list entirely: **recQ** (PP_4516), **recN** (PP_4729), **recX** (PP_1630, immediately adjacent to recA), **radA/sms** (PP_4644), the **sbcCD** nuclease pair (PP_2024/PP_2025), and the natural-transformation loader **dprA** (PP_0069).

A crucial strain-direct caveat qualifies the "covered" verdict: despite an apparently complete gene complement, KT2440 is experimentally a **poor homologous recombiner with a weak SOS response** (PMID: 33393180). Gene presence therefore does **not** equal high recombination proficiency in this organism, and this should be flagged for any downstream phenotype-linked curation. Finally, the replication-restart primosome components **priB** and **dnaT** appear genuinely absent from the genome, consistent with the *Pseudomonas* PriA–PriC restart lineage; these steps should be marked `not_expected_in_target_taxon` rather than `gap`.

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2. Target-Organism Pathway Definition

What the module encompasses

The `recfor_recombination` module represents the **RecF(OR) pathway of homologous recombination and single-stranded-DNA (ssDNA) gap repair**. Mechanistically this is the presynaptic-through-resolution process by which a recombinogenic RecA nucleoprotein filament is assembled on ssDNA gaps (as opposed to double-strand-break ends, which are the preferred substrate of the RecBCD pathway), followed by RecA-mediated homology search and strand invasion, branch migration, and Holliday-junction resolution. In *E. coli* this pathway "repairs such ssDNA gaps by processing them to produce a recombinogenic RecA nucleofilament during the presynaptic phase" (PMID: 35653392).

The presynaptic phase specifically involves: (i) resection/extension of the ssDNA gap by the 5′→3′ exonuclease **RecJ** and the 3′→5′ helicase **RecQ**; (ii) protection of exposed ssDNA by **SSB**; and (iii) **RecFOR**-mediated displacement of SSB and loading of **RecA**. Impairing "either the extension of the ssDNA gap (mediated by the nuclease RecJ and the helicase RecQ) or the loading of RecA (mediated by RecFOR) leads to a decrease in" homology-directed gap repair (PMID: 35653392). This is the biochemical scope that defines module membership and justifies inclusion of recQ and recJ even though the KEGG metadata files them under other buckets.

Neighboring pathways that must be kept separate

For curation purposes the module boundary must exclude several adjacent processes whose genes bleed into KEGG `map03440`:

Neighboring process	KEGG map	Genes drawn in erroneously
DNA replication (Pol III holoenzyme, Pol I)	ppu03030	dnaN, holC, dnaEA, holB, dnaQ, dnaX, holA, ssb, PP_0353, PP_4768
Base-excision / mismatch repair	ppu03410	recJ (mis-bucketed)
Nucleotide-excision repair	ppu03420	polA
Non-homologous end joining	— (absent in KT2440)	not applicable

The Pol III holoenzyme subunits appear in `map03440` only because KEGG renders the same replisome cartoon in both maps; their presence in the candidate list is a **rendering artifact, not a recombination assignment**. Note that **ssb** is genuinely dual-role (it is required in both replication and RecFOR presynapsis) — it should be retained as a shared/accessory member rather than deleted.

Alternate names and database definitions

The module is variously called the **RecF pathway**, **RecFOR pathway**, or **RecF(OR) recombination**. KEGG groups it under the umbrella map "Homologous recombination" (`map03440` / organism-specific `ppu03440`), which conflates the RecFOR and RecBCD sub-pathways into one map. The genus-level literature is explicit that these are two distinct routes: "Two pathways are responsible for homologous recombination in *Pseudomonas aeruginosa*: the RecBCD pathway and the RecFOR pathway" (PMID: 29633970). Curators should treat `recfor_recombination` as the RecFOR-specific sub-module and, if a separate RecBCD module exists, cross-reference rather than merge.

3. Expected Step Model

The following step model represents the RecFOR pathway as expected in a Gammaproteobacterium such as KT2440, annotated with the KT2440 locus that satisfies each step.

```
ssDNA gap (e.g., behind a stalled/reprimed replication fork)
      |
      v
[1] Gap resection / extension
      RecJ 5'->3' exonuclease  ...... PP_1477 (recJ)      COVERED
      RecQ 3'->5' helicase     ...... PP_4516 (recQ)      COVERED (missing from list)
      |
      v
[2] ssDNA protection
      SSB                      ...... PP_0485 (ssb)       COVERED (dual-role)
      |
      v
[3] Mediator / RecA loading (presynapsis)
      RecFOR complex:
        RecF                   ...... PP_0012 (recF)      COVERED
        RecO                   ...... PP_1435 (recO)      COVERED
        RecR                   ...... PP_4267 (recR)      COVERED
      |
      v
[4] Synapsis / strand exchange
      RecA recombinase         ...... PP_1629 (recA)      COVERED
      RecX (RecA modulator)    ...... PP_1630 (recX)      COVERED (missing from list)
      RadA/Sms (branch migr.)  ...... PP_4644 (radA)      COVERED (missing from list)
      |
      v
[5] Branch migration + resolution
      RuvA                     ...... PP_1216 (ruvA)      COVERED
      RuvB                     ...... PP_1217 (ruvB)      COVERED
      RuvC                     ...... PP_1215 (ruvC)      COVERED
      RecG (alt. migration)    ...... PP_5310 (recG)      COVERED
      |
      v
[6] Replication restart
      PriA                     ...... PP_5088 (priA)      COVERED
      PriB                     ...... ABSENT              NOT_EXPECTED
      DnaT                     ...... ABSENT              NOT_EXPECTED
      (PriA-PriC/DnaC-independent restart lineage)

Parallel presynaptic route (double-strand-break ends):
      RecBCD                   ...... PP_4673/PP_4674/PP_4672   COVERED
Accessory / overlapping:
      RecN (DSB cohesion/SMC)  ...... PP_4729 (recN)     COVERED (missing from list)
      SbcCD (Mre11/Rad50-like) ...... PP_2024/PP_2025    COVERED (missing from list)
      DprA (transformation)    ...... PP_0069 (dprA)     COVERED (missing from list)
```

Every catalytic step of the RecFOR pathway is encoded in KT2440. The only genuinely absent components are the accessory primosome loaders PriB and DnaT, which are lineage-specific losses and should not be scored as gaps.

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4. Candidate Genes and Evidence

4.1 High-confidence RecFOR core (retain, covered)

Gene	Locus	UniProt	KO	Role	Evidence type
recF	PP_0012	Q88RW7	K03629	RecA-loading mediator	Homology + KEGG KO; strong
recO	PP_1435	Q88MY3	K03584	SSB displacement, RecA loading	Homology + genus structure
recR	PP_4267	Q88F32	K06187	Forms RecF/RecO complexes	Genus structural (P. aeruginosa)
recA	PP_1629	Q88ME4	K03553	Strand-exchange recombinase	Strain-direct (KT2440 SOS study)
recJ	PP_1477	Q88MU1	K07462	5′→3′ ssDNA exonuclease	Homology; mechanistic
recQ	PP_4516	—	K03654	3′→5′ helicase, gap extension	Homology; **not in candidate list**
ssb	PP_0485	Q88QK5	K03111	ssDNA protection (dual-role)	Homology; shared with replication

Curation notes. RecF, RecO and RecR are the defining triad; the *P. aeruginosa* RecR crystal structure confirms that in this genus RecR forms RecF/RecO complexes that load RecA onto ssDNA (PMID: 29633970), a strong genus-level transfer to KT2440. RecA is the one component with **direct KT2440 experimental characterization**, via the SOS-response study (PMID: 33393180). RecJ carries a KEGG primary bucket of `ppu03410` (base-excision/mismatch repair) and recQ is absent from the list entirely, yet both are integral RecFOR presynaptic factors per the mechanistic literature (PMID: 35653392); they must be **added to / retained in** the module.

4.2 Downstream resolution machinery (retain, covered)

Gene	Locus	Role
ruvA	PP_1216	Holliday-junction branch-migration subunit
ruvB	PP_1217	Holliday-junction branch-migration ATPase
ruvC	PP_1215	Holliday-junction resolvase (EC 3.1.21.10)
recG	PP_5310	Alternative branch-migration helicase

These form a contiguous ruvCAB locus (PP_1215–PP_1217) plus recG, and are unambiguous by homology. Retain as covered.

4.3 RecBCD parallel pathway (retain, covered but separate sub-pathway)

Gene	Locus	Role
recB	PP_4673	Helicase/nuclease subunit (EC 3.1.11.5 / 5.6.2.4)
recC	PP_4674	Recognition subunit
recD	PP_4672	Helicase subunit (EC 5.6.2.3)

Contiguous recCBD operon. These serve the DSB-end presynaptic route, parallel to RecFOR (PMID: 29633970). Keep them associated with the HR map but flagged as **RecBCD sub-pathway**, not RecFOR proper.

4.4 Replication genes to REMOVE from the RecFOR module (over-annotation)

Gene	Locus	Primary bucket	Why remove
dnaN	PP_0011	ppu03030	β-clamp; replication processivity
polA	PP_0123	ppu03420	DNA Pol I; NER/gap filling
PP_0353	PP_0353	ppu03030	generic "Exonuclease"
holC	PP_0979	ppu03030	Pol III χ subunit
dnaEA	PP_1606	ppu03030	Pol III α subunit
holB	PP_1966	ppu03030	Pol III δ′ subunit
dnaQ	PP_4141	ppu03030	Pol III ε subunit
dnaX	PP_4269	ppu03030	Pol III γ/τ subunit
PP_4768	PP_4768	ppu03030	generic "Exonuclease"
holA	PP_4796	ppu03030	Pol III δ subunit

These ten loci are replisome/repair-polymerase components that KEGG renders in `map03440` because the same replisome cartoon is shared between the replication and recombination maps. They should be **removed from the RecFOR module** (they belong to the replication module), while noting that the replisome is of course biochemically required to complete recombination-associated DNA synthesis.

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5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Missing-from-metadata but present-in-genome (add to module)

The candidate list omits several genome-encoded HR genes that were located directly via KEGG KO→gene mapping:

- **recQ** = **PP_4516** (K03654, "ATP-dependent DNA 3′-5′ helicase") — an integral presynaptic factor; its omission is the most consequential gap in the list.
- **recN** = **PP_4729** (K03631) — SMC-family DSB-repair cohesion factor.

- **recX** = **PP_1630** (K03565) — RecA modulator, located immediately adjacent to recA (PP_1629), a strong syntenic argument for functional association.
- **radA/sms** = **PP_4644** (K04485) — branch-migration/recombination accessory.
- **sbcC** = **PP_2024** (K03546) + **sbcD** = **PP_2025** (K03547) — Mre11/Rad50-like nuclease pair involved in end processing.
- **dprA** = **PP_0069** (K04096) — natural-transformation RecA loader.

All definitions were retrieved directly from the KEGG REST list endpoints. These are candidates to be **added** to the module as covered (recQ, recX) or accessory/uncertain (recN, radA, sbcCD, dprA).

5.2 Genuinely absent (mark not_expected_in_target_taxon)

- **priB** (K02686) and **dnaT** (K02317) — KEGG name and KO searches (`find/ppu/priB`, `find/ppu/dnaT`, `link ko:K02686`, `link ko:K02317`) returned no KT2440 gene. Their absence is consistent with a *Pseudomonas* **PriA–PriC / DnaC-independent replication-restart lineage**. PriA itself (PP_5088) is present. Mark PriB and DnaT `not_expected_in_target_taxon`, not `gap`.

5.3 Over-annotation / broad mappings

- The ten Pol III / Pol I / generic-exonuclease loci (Section 4.4) are the primary over-propagation problem.
- **PP_0353** and **PP_4768** are annotated only as generic "Exonuclease" — broad, non-specific mappings that should not anchor a recombination step.
- **ssb (PP_0485)** carries a replication primary bucket but is legitimately dual-role; keep it, but annotate the dual assignment.
- The broad EC numbers on helicase/nuclease subunits (e.g., recB EC 3.1.11.5 / 5.6.2.4; priA/recG EC 5.6.2.4) reflect enzyme-family generalizations and should not be over-interpreted as evidence of a specific recombination substrate.

5.4 Phenotype caveat (strain-direct)

Even with a complete gene complement, "*Pseudomonas putida* strain KT2440 is very sensitive to DNA damage and displays poor homologous recombination efficiencies" (PMID: 33393180). This inefficiency is attributed to a faulty RecA–LexA SOS interplay. Curation should record that

module satisfiability (gene presence) is decoupled from measured recombination proficiency in this strain — a distinction that matters if the module is ever tied to functional phenotype scoring.

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6. Module and GO-Curation Recommendations

Overall module verdict: `module_needs_revision` — the pathway is `covered`, but the gene set is wrong at both boundaries.

Recommended per-step scoring:

Step / component	Locus	Recommended status
RecF	PP_0012	covered
RecO	PP_1435	covered
RecR	PP_4267	covered
RecA	PP_1629	covered (strain-direct)
RecJ (presynaptic)	PP_1477	covered — reassign from ppu03410 bucket
RecQ (presynaptic)	PP_4516	covered — **add to module**
SSB	PP_0485	covered (dual-role, retain)
RuvABC	PP_1215/16/17	covered
RecG	PP_5310	covered
RecBCD	PP_4672/73/74	covered (RecBCD sub-pathway)
PriA	PP_5088	covered
RecX	PP_1630	candidate — add (adjacent to recA)
RecN	PP_4729	candidate_uncertain — add
RadA/Sms	PP_4644	candidate_uncertain — add
SbcCD	PP_2024/25	candidate_uncertain — add
DprA	PP_0069	candidate_uncertain — add
PriB	absent	not_expected_in_target_taxon
DnaT	absent	not_expected_in_target_taxon
Pol III subunits (dnaN, holABC, dnaEA, dnaQ, dnaX)	various	remove — belongs to ppu03030 replication
polA	PP_0123	remove — belongs to ppu03420 NER
Generic exonucleases (PP_0353, PP_4768)	—	remove — non-specific

Boundary correction. The generic KEGG `map03440` boundary is **wrong for module curation** because it merges RecFOR and RecBCD and pulls in the replisome. Recommend splitting into (a) a RecFOR presynaptic module and (b) a RecBCD module, with shared downstream RuvABC/RecG resolution.

GO-curation. No new GO term requests appear strictly necessary; existing terms cover the process: - GO:0000730 (DNA recombinase assembly) / GO:0000731 (DNA synthesis involved in DNA repair) for RecFOR loading, - GO:0000724 (DSB repair via homologous recombination), - GO:0009432 (SOS response) for the RecA/LexA caveat.

A curation note requesting that **RecQ and RecJ be annotated to the RecFOR/HR process** (rather than only to their current excision-repair buckets) would improve consistency.

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7. Genes to Promote to Full `fetch-gene` Review

Priority order for full individual-gene review:

1. **recQ (PP_4516)** — highest priority; missing from the list, integral presynaptic helicase, needs explicit module assignment and confirmation of KO K03654 identity.
2. **recX (PP_1630)** — adjacent to recA; confirm RecA-modulator role and whether it should be covered vs. accessory.
3. **recA (PP_1629)** — only strain-direct-characterized gene; promote to anchor the phenotype caveat (poor HR, weak SOS) with primary literature.
4. **recN (PP_4729), radA/sms (PP_4644), sbcC/sbcD (PP_2024/PP_2025), dprA (PP_0069)** — accessory HR genes absent from the list; review to decide covered vs. candidate_uncertain.
5. **recJ (PP_1477)** — resolve the bucket conflict (currently ppu03410) and formally reassign to the RecFOR presynaptic step.
6. **priB / dnaT** — confirm true genomic absence (negative result) to lock in `not_expected_in_target_taxon`.

The ten replication-polymerase loci do **not** warrant HR-module fetch-gene review; they should simply be removed and left to the replication module.

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8. Evidence Base and Key References

PMID	Title (abbrev.)	How it supports the review
33393180	*The faulty SOS response of P. putida KT2440 stems from an inefficient RecA-LexA interplay*	**Strain-direct.** Establishes that KT2440 recombines poorly and has a weak SOS response despite a complete gene set — the central phenotype caveat.
35653392	*Single strand gap repair: the presynaptic phase...*	Defines the RecF-pathway presynaptic step and shows RecJ + RecQ (gap extension) and RecFOR (RecA loading) are integral — justifies module scope and inclusion of recQ/recJ.
29633970	*Crystal structure of RecR... from P. aeruginosa PAO1*	**Genus-level.** States RecBCD and RecFOR are the two HR pathways in *Pseudomonas* and that RecR forms RecF/RecO complexes loading RecA — supports the parallel-pathway expected model.
26195593	*RecF and RecR play critical roles in HR and SSA in mycobacteria*	Cross-taxon support that RecF/RecR are central mediators of RecA loading; RecR essential for all HR — reinforces core-triad importance (weak transfer, different phylum).
37070184	*RecA and SSB genome-wide distribution in ssDNA gaps... in E. coli*	Genome-scale confirmation that RecA/SSB coat ssDNA gaps and that RecBCD- and RecFOR-independent RecA loading pathways exist — informs accessory-factor scope.
36985274	*Chromosome segregation and cell division defects...*	Context that HR repairs both DSBs and ssDNA gaps, linking recombination to genome maintenance.
32297860	*Single-molecule observation of ATP-independent SSB displacement by RecO*	Mechanistic support for RecO's SSB-displacement role during RecA loading.

Evidence quality summary. Only RecA has *direct experimental characterization in KT2440* (PMID: 33393180). The RecFOR triad mechanism transfers **strongly** from the genus (*P. aeruginosa*, PMID: 29633970) and **strongly** from the well-characterized *E. coli* RecF pathway (PMID: 35653392). Gene identities (KO/locus assignments, presence/absence of recQ, recN, recX, radA, sbcCD, dprA, priB, dnaT) rest on **KEGG REST homology mapping**, not on KT2440 experiments, and should be treated as high-confidence-by-homology but not experimentally proven.

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9. Limitations and Knowledge Gaps

- **Homology-based gene calls.** Presence/absence of recQ, recN, recX, radA, sbcCD, dprA, priB and dnaT was determined by KEGG KO→locus mapping and name searches, not by experimental function assays in KT2440. A false-negative in a KEGG name search could in principle mask a divergent priB/dnaT ortholog.
- **Genotype–phenotype decoupling.** The "covered" verdict is a gene-presence statement. The strain's documented poor HR efficiency (PMID: 33393180) means module satisfiability does not predict recombination proficiency.
- **RecFOR vs RecBCD attribution.** KEGG merges the two pathways; the exact division of labor (which route dominates ssDNA-gap vs DSB repair) has not been measured in KT2440 specifically.
- **Accessory-factor roles.** RecN, RadA, SbcCD and DprA are placed by homology; their quantitative contribution to HR in KT2440 is unknown.
- **No expression/proteomics evidence** was incorporated; all conclusions are sequence/pathway-database derived plus targeted literature.

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10. Proposed Follow-up Actions

1. **Revise the module gene set**: remove the ten Pol III/Pol I/generic-exonuclease loci; add recQ (PP_4516), recX (PP_1630), and (as accessory) recN, radA, sbcCD, dprA; retain ssb as dual-role; reassign recJ to the RecFOR presynaptic step.
2. **Mark priB and dnaT `not_expected_in_target_taxon`** with a note documenting the *Pseudomonas* PriA–PriC restart lineage; verify by a dedicated ortholog search (e.g., HMM against PriB/DnaT profiles) to convert the negative KEGG search into a confident absence call.
3. **Promote recQ, recX, recA, recJ, and the accessory quartet (recN/radA/sbcCD/dprA) to full `fetch-gene` review** (Section 7 order).

4. **Split the generic HR module** into RecFOR and RecBCD sub-modules sharing downstream RuvABC/RecG, and correct the KEGG-inherited boundary.
5. **Record the phenotype caveat** (PMID: 33393180) in the module annotation so that gene-presence coverage is not misread as high recombination competence.
6. **Optional experimental resolution** (expert questions): (a) measure relative RecFOR vs RecBCD contributions to gap vs DSB repair in KT2440; (b) test whether recQ/recJ deletion phenocopies RecFOR deletion; (c) confirm the SOS/RecA-LexA defect's impact on module-associated repair phenotypes.

Prepared for manual module-satisfiability and gene-annotation curation. Verdict: pathway `covered`, gene set `module_needs_revision`. All locus and KO identifiers verified via KEGG REST; strain-direct evidence limited to RecA/SOS (PMID: 33393180); mechanistic scope from E. coli/ Pseudomonas (PMID: 35653392, PMID: 29633970).
